

EVALUATION AND VALIDATION OF THE MODIS LAI ALGORITHM WITH DIGITAL HEMISPHERICAL PHOTOGRAPHY AT BHITAR KANIKA MANGROVE FOREST, INDIA

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ABSTRACT

Estimation of LAI is really a challenge and researchers are developing new methods to get more accurate estimation of LAI. MODIS continuously provides globally temporal LAI product and it is easily accessible for application and development purpose. But main problem is the spatial resolution (500m) where the main target features has affected due to the presence of unwanted features like clouds and others. It was concluded by previous researchers that MODIS MOD15A2H products give the overestimation or underestimation of LAI. From the Choudhury's method [1] of view, using the available MODIS products like vegetation indices, view zenith angle, sun azimuthal angle information LAI estimation can be improved. In this study an algorithm was developed on the basis of this Choudhury method to extract the improved LAI estimation by indirectly retrieving of canopy leaf angle distribution function, extinction coefficient from the MODIS MODIS13Q1 product. This modified Choudhury method (ModcModLAIE) applied in the Bhitar Kanika mangrove forest of Odisha, India for estimating the LAI. It was proved and validated with the help of Digital Hemispherical Photography (DHP) that from the available MODIS products itself it is possible to extract finer resolution (250m) improved LAI product. DHP analysis works on the basic principle of Beer's law, gap fraction calculation. At the time of comparison, it has shown that DHP extracted LAI has the higher correlation with ModcModLAIE output than the MOD15A2H LAI product. So, this research has proved the overestimation and underestimation fact of MODIS LAI product as well as it has shown a way to extract an improved LAI estimation from available course resolution MODIS products.

Index Terms— MODIS, DHP, ModcModLAIE, gap fraction, Beer's law, LAI

1. INTRODUCTION

Terrestrial vegetation has a great important role in the land-atmosphere-ocean coupled system as they maintain the energy, water, nutrient circulation in a balanced way. Leaf Area index (LAI), a key biophysical parameter for plant physiologist, modeler to estimate foliage cover, biomass estimation, crop yield production, as well as monitoring and forecasting of crop growth [2]. It is the ratio of one side leaf area per unit ground surface area. This LAI can relate to the CO₂-H₂O exchange which is the vital part of photosynthesis, evapotranspiration (ET) [3]. It can serve as the indicator of forest stress and health [4]. There are mainly two methods to derive in-situ LAI: direct and indirect [5]. Indirect methods using optical instruments such as LI-COR LAI-2000, TRAC or DHP have been widely used for the estimation of LAI [5]. Among them, DHP has the advantage of providing a permanent 2-dimensional record of canopy structure [6] and also cost effective than plant canopy analyzer. Remotely sensed data has a great potential to retrieve of LAI, Madugundu et al. [7] estimated LAI using LISS-IV multispectral data which was based on the regression analysis between ground based observation and positively correlated NDVI [8] in the Western Ghats of Karnataka, India. Chhabra et al. [9] presented a detailed spatio-temporal and inter-seasonal analysis of LAI patterns in different forest types of India using MODIS 8-day temporal global LAI/ fPAR product for the year 2005 at 1-km spatial resolution where the range of estimated LAI varied from 0.1-6.9 among the different forest types.

According to the previous researches some research gaps have found to further improvement in LAI estimation from MODIS data product itself and with that following objectives have taken into account,

- 1) Mapping of LAI from MODIS (MOD15A2H) open source data products of Bhitarkanika mangrove forest, Odisha, India.
- 2) Modification of Choudhury method [1] to estimate more refined LAI from available MODIS products.
- 3) Comparison and validation of above stated goals with DHP extracted LAI.

2. MATERIALS AND METHODS

2.1. Study area

Bhitarkanika wildlife sanctuary is a rich, green vibrant eco-system lying on the estuarine region of Brahmani-Baitarani of Kendrapara district of Odisha. It is surrounded by the Bay of Bengal in the east, village of Kendrapara district in west, Baitarani and Dharma river in the north and Mahanadi delta in south. It covers an area of 672 sq. km of mangrove forest in between 86°48'–87°03' E longitude and 20°33'–20°47' N latitude with an elevation between 1.5 to 2 meters above mean sea level. The average rainfall is about 1642 mm.

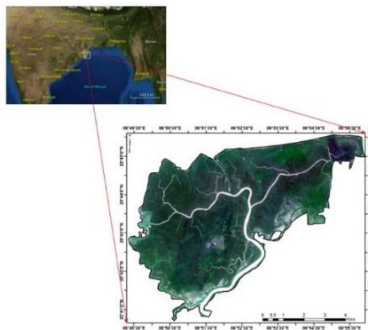


Figure 1. Bhitarkanika (Mangrove forest) national sanctuary, Odisha, India

Mean relative humidity ranges from 70 to 85% throughout the year where maximum and the minimum temperature is 41°C and 9°C during May and January respectively. Bhitarkanika is home to over 55 species of mangroves, whereas the main dominant species are *Avicennia officinalis*, *Ceriops decandra*, *Cynometra iripa*, *Exoceria agallocha*, *Heritiera fomes*, *Sonneratia apetala*, *Bruguiera sexangula*, *Xylocarpus mekongensis*, *Aglaia cucullata*, *Pongamia pinnata* etc.

The overall methodology has shown by following flowchart,

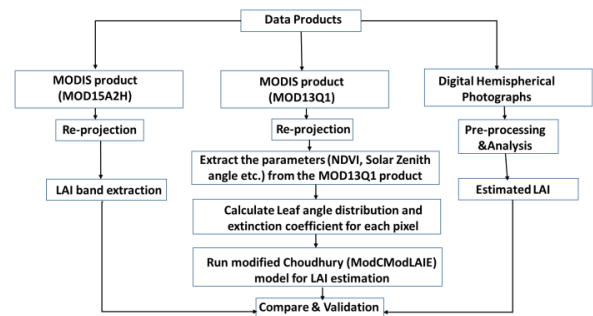


Figure 2. Flowchart showing the methodology of the study

2.2. Data and other resources

2.2.1. Spatial Data

MOD13Q1 and MOD15A2H in 16 and 8-days interval respectively from October, 2018 to December, 2018

2.2.2. Ground sample

Hemispherical photographs were taken in December-2018 by Nikon D5300 with Nikon AF-S Fisheye NIKKOR 8-15mm f/3.5-4.5E ED Lens. 20mX20m plot was selected and followed a field architecture, from there 13 photographs were taken per plots.

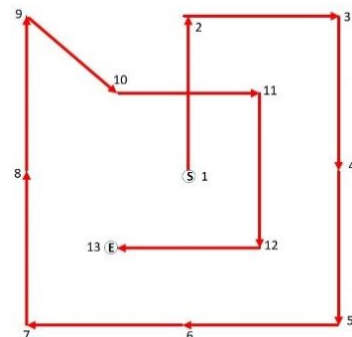


Figure 3. Field plot design for sample (DHP) collection

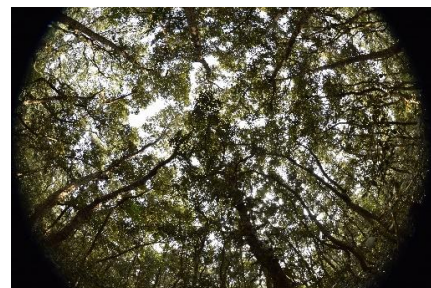


Figure 4. Hemispherical photograph collected from a plot with the fish eye lens

2.2.3. Software

Python, Arc Map-10.3, CAN-EYE (for DHP processing and analysis), MODIS Reprojection Tool (MRT)

2.3. Methods

This Choudhury method is based on the vegetation indices mainly, NDVI. He calculated the fractional vegetation cover (f) using the following equation,

$$f = 1 - \ln\left\{\frac{NDVI_{max} - NDVI}{NDVI_{max} - NDVI_{min}}\right\}^{1/\varepsilon}$$

$$LAI = 1 - \ln\left\{\frac{(1-f)}{-\beta}\right\}$$

The parameter ε is a function of canopy leaf angle distribution with values near 1.4 for erectophile canopies and near 0.8 for planophile canopies [1]. Where β is a function of extinction that ranges from 0.42 to 0.91 [1].

For ModcModLAIE; NDVI, sun view angle, solar zenith angle these three bands were extracted from MOD13Q1 data products. From that parameters leaf angle distribution function, extinction function was calculated and finally all these extracted parameters were used in Choudhury's method to get the final LAI.

DHP image processing for LAI estimation works on the principle of Beer Lambert's equation,

$$P_0 = \exp\left[-\int_0^s k \, ds\right]$$

where P_0 is the probability of light passing through the medium, s is the path through the medium, and k is the extinction factor. Some modified version of this equation was developed for gap fraction calculation [10]-[11][12], and this gap fraction measurement helps to estimate LAI.

The CAN_EYE software proposes an estimation of the LAI derived from the equation, called LAI57. CAN_EYE also proposed other ways of computing PAI using Miller's formula [10],

$$PAI = 2 \int_0^{\pi/2} [-\ln(P_0(\theta_v) \cos(\theta_v) \sin(\theta_v))] \, d\theta_v$$

Where θ_v is the view zenith angle.

3. RESULTS

From the MOD15A2H product LAI band was extracted and after reprojection with MRT tool the LAI map (Figure. 5) was generated. In this case it showed that the LAI values mostly ranged between 4-5.

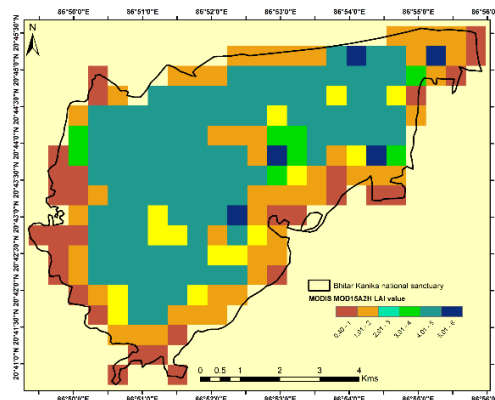


Figure 5. LAI map from MODIS MOD15A2H LAI product

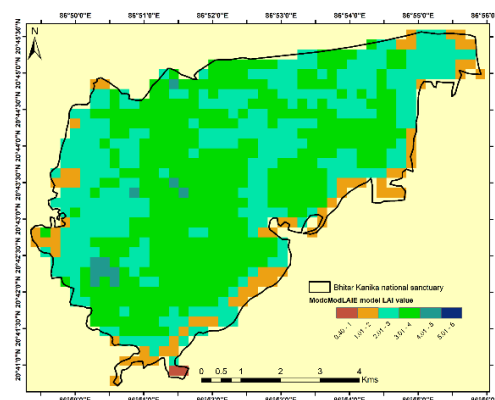


Figure 6. LAI map from modified Choudhury model (ModcModLAIE)

Utilizing the NDVI, solar zenith angle, sun view angle bands of MODIS MOD13Q1 data product the LAI map (Figure. 6) was generated through modified Choudhury model where the LAI values mostly ranged between 2-4.

CAN_EYE software calculates the LAI using gap fraction technique, this process is done after gamma correction and classification of vegetation in one class and remaining are in second class.

MOD15A2H product shows sometimes some arbitrary LAI value (in the range of 25-50) or even zero which should not be possible. It was found that in a pixel of

MODIS LAI product had the LAI value zero whereas for the same location and same date NDVI product had a value of 0.6, but it should not be logical; that means it may be an error or noise for that pixel. whereas, the developed model didn't show such an error.

The ModcModLAIE model output was validated with DHP analyzed LAI of 15 ground sample plot, which showed 67.4% of correlation whereas, MOD15A2H LAI data products showed 44% correlation.

4. CONCLUSION

Choudhury model had some disadvantages like, they used only one generalized extinction coefficient value and canopy leaf angle distribution function for whole but it should be changed according to the canopy architecture, solar view angle, and solar zenith angle. Considering all these Choudhury model was modified for LAI estimation which has advantages like, easily able to get the better spatial resolution (250m) of LAI product for different canopy leaf angle distribution function and extinction coefficients with better accuracy with respect to indirect in-situ ground sampling. Although, only 15 plots of hemispherical photographs were analyzed, it showed higher accuracy with ModcModLAIE model output than the MODIS LAI products. So, this modified Choudhury model deserved the potential to improve LAI estimation from available coarse resolution MODIS products. Further this modified model will be tested for varied range of DHP processed LAI for the whole study area.

11. REFERENCES

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